

CLINISCAN

Lung-Abnormality Detection on Chest X-Rays Using AI

INFOSYS SPRINGBOARD

Milestone 3: Model Optimization & Refinement

Submitted By: Ala Agastien

1. INTRODUCTION

The objective of Milestone 3 is to significantly improve upon the baseline models established in Milestone 2 by applying systematic optimization techniques. This phase focuses on hyperparameter tuning, advanced data augmentation, staged transfer learning, Grad-CAM interpretability, and a properly structured train/validation dataset split for the YOLO detection model.

The key improvements implemented in this milestone are:

- **Advanced Data Augmentation:** Albumentations pipeline with horizontal flips, ShiftScaleRotate, brightness/contrast adjustment, CLAHE, Gaussian noise, and CoarseDropout.
- **Staged Fine-Tuning (ResNet50):** Two-phase training — frozen backbone (head only) followed by full model fine-tuning with cosine annealing and label smoothing.
- **Improved Architecture:** ResNet50 classifier head upgraded to a two-layer MLP with dropout regularization.
- **Proper Train/Val Split (YOLO):** Dataset split into 80% train / 20% validation for unbiased detection evaluation.
- **Grad-CAM Visualization:** Gradient-weighted Class Activation Maps generated to visualize model attention on lung regions.
- **Comprehensive Evaluation:** Per-class reports, confusion matrices, and final summary metrics for both models.

2. DATA DESCRIPTION

The system continues to utilize the VinDr-CXR dataset with 15,000 annotated chest X-ray training images. In this milestone, the data pipeline was refined with stricter train/validation splits and richer augmentation strategies.

2.1 Key Data Characteristics

- **Annotations:** train.csv with disease class IDs and bounding box coordinates (x_min, y_min, x_max, y_max).
- **Metadata:** train_meta.csv providing original image dimensions (dim0, dim1) per image ID.
- **Image Format:** PNG images at native resolution; resized to 224×224 for classification and 640×640 for detection.
- **Classes:** 14 disease categories (class IDs 0–13); class ID 14 (No Finding) excluded from training.

3. DATA PREPROCESSING

3.1 Classification Preprocessing

For the ResNet50 classification model, the preprocessing pipeline was enhanced as follows:

- **Filtering:** Rows with `class_id == 14` (No Finding) were removed. A single primary label per image was assigned using `groupby().first()` to maintain a clean single-label mapping.
- **Train/Val Split:** An 80/20 stratified split was applied using `sklearn.model_selection.train_test_split` to ensure balanced class representation across splits.
- **Normalization:** ImageNet mean [0.485, 0.456, 0.406] and std [0.229, 0.224, 0.225] applied for ResNet50 compatibility.

3.2 Advanced Augmentation Pipeline (Albumentations)

A comprehensive augmentation pipeline was implemented using the Albumentations library for the training set:

Augmentation	Parameters	Probability
Resize	224 × 224	Always
HorizontalFlip	—	p = 0.5
ShiftScaleRotate	shift=0.05, scale=0.1, rotate=15°	p = 0.4
RandomBrightnessContrast	brightness=0.2, contrast=0.2	p = 0.4
CLAHE	clip_limit=2.0, tile_grid=(8,8)	p = 0.3
GaussNoise	var_limit=(10.0, 50.0)	p = 0.2
CoarseDropout	max_holes=8, size=16×16	p = 0.2
Normalize	ImageNet mean/std	Always

Validation transforms were limited to Resize and Normalize only to avoid data leakage.

3.3 YOLO Detection Preprocessing

- **Train/Val Split:** 80/20 split at the image level before building the YOLO dataset directories.
- **YOLO Format Conversion:** Bounding box coordinates normalized to (x_center, y_center, width, height) relative to image dimensions.
- **Empty Labels:** Empty .txt label files generated for images with no detected abnormalities.
- **Dataset YAML:** `data_m3.yaml` configured with correct train/val paths and 14 class names.

4. CLASSIFICATION MODEL (ResNet50 — Staged Fine-Tuning)

4.1 Model Architecture

The ResNet50 backbone pretrained on ImageNet was retained. The final fully-connected layer was replaced with an enhanced two-layer MLP head:

Layer	Details
Backbone	ResNet50 (ImageNet pretrained)
Dropout	rate = 0.4
Linear	in_features → 256
ReLU	Activation
Dropout	rate = 0.2
Linear	256 → 14 (num classes)

Total trainable parameters: ~25.6 million (full model).

4.2 Staged Training Strategy

Phase 1 — Frozen Backbone (5 Epochs)

- **Goal:** Train only the new classification head while the ResNet50 backbone remains frozen.
- **Optimizer:** Adam with $lr = 1e-3$.
- **Loss:** CrossEntropyLoss with label smoothing = 0.1.
- **Rationale:** Allows the new head to adapt to the medical domain before full backpropagation.

Phase 2 — Full Fine-Tuning (10 Epochs)

- **Goal:** Unfreeze all layers and fine-tune end-to-end at a much lower learning rate.
- **Optimizer:** AdamW with $lr = 1e-5$, `weight_decay = 1e-4`.
- **Scheduler:** CosineAnnealingLR with `T_max = 10`, `eta_min = 1e-7`.
- **Checkpoint:** Best model saved as `cliniscan_best_cls_m3.pth` based on validation F1 score.

4.3 Classification Results

Metric	Baseline (M2)	Optimized (M3)
--------	---------------	----------------

Validation Accuracy	~Baseline	Improved
Val F1 (Macro)	~Baseline	Best checkpoint saved
Val AUC (OvR)	~Baseline	Multi-class OvR AUC
Training Phases	Single phase (5 epochs)	Two-phase (5 + 10 epochs)
Augmentation	Resize only	Full Albumentations pipeline
Head Architecture	Single linear layer	Two-layer MLP with dropout

The per-class classification report was generated covering all 14 disease categories including Aortic enlargement, Atelectasis, Calcification, Cardiomegaly, Consolidation, ILD, Infiltration, Lung Opacity, Nodule/Mass, Other lesion, Pleural effusion, Pleural thickening, Pneumothorax, and Pulmonary fibrosis.

5. GRAD-CAM VISUALIZATION

Gradient-weighted Class Activation Mapping (Grad-CAM) was implemented for the ResNet50 classifier to provide clinical interpretability — visualizing which regions of the chest X-ray the model attends to when making predictions.

5.1 Implementation Details

- **Target Layer:** model.layer4 (final convolutional block of ResNet50).
- **Forward Hook:** Captures feature map activations at the target layer.
- **Backward Hook:** Captures gradients flowing back through the target layer.
- **Heatmap Generation:** Activations are globally average-pooled by their gradient weights and ReLU is applied to retain positive influences.
- **Overlay:** Heatmap is upsampled to input image size and blended with the original X-ray using a colormap overlay (alpha=0.5).

5.2 Grad-CAM Visualization Output

For 6 random validation samples, each visualization shows three panels:

- Original chest X-ray image (denormalized).
- Grad-CAM heatmap (jet colormap showing high-activation regions).
- Overlay of the heatmap on the original image for clinical reference.

The Grad-CAM outputs confirm that the model attends to anatomically relevant lung regions for most disease categories, supporting clinical trust in the model's predictions.

6. DETECTION MODEL (YOLOv11s — Tuned Hyperparameters)

6.1 Architectural Choice

The YOLOv11s (small) architecture from Ultralytics was retained from Milestone 2 but retrained with significantly improved hyperparameters and a proper train/validation split.

6.2 Training Configuration

Parameter	Milestone 2 Value	Milestone 3 Value
Image Size (imgsz)	1024×1024	640×640
Epochs	25	30
Batch Size	8	16
Learning Rate (lr0)	Default	0.005
LR Final (lrf)	Default	0.01
Momentum	Default	0.937
Weight Decay	Default	5e-4
Warmup Epochs	Default	3
Cosine LR Schedule	No	Yes
Mosaic Augmentation	Default	1.0
MixUp Augmentation	Default	0.15
Rotation (degrees)	Default	10.0
Val Split	Train = Val	80% train / 20% val

6.3 Detection Evaluation Metrics

The trained YOLOv11s model was evaluated on the held-out validation set using the following metrics:

Metric	Description	Result
mAP@0.50	Mean Average Precision at IoU threshold 0.50	Reported
mAP@0.50:0.95	Mean AP averaged across IoU thresholds 0.50–0.95	Reported

Precision (mp)	Mean precision across all classes	Reported
Recall (mr)	Mean recall across all classes	Reported

Bounding box visualizations were generated for validation samples to confirm spatial localization of detected abnormalities. Each detection is color-coded per disease class with confidence scores displayed.

7. SYSTEM WORKFLOW & MODEL SAVING

The CliniScan integrated pipeline in Milestone 3 follows an enhanced two-stage workflow:

Stage	Component	Output
Input	Chest X-ray PNG image	Raw image tensor
Preprocessing	Albumentations pipeline (val: resize + normalize)	Normalized tensor 224×224
Detection	YOLOv11s (best.pt)	Bounding boxes + class + confidence
Classification	ResNet50 + MLP head (best checkpoint)	14-class probability distribution
Interpretability	Grad-CAM on ResNet50.layer4	Heatmap overlay on X-ray
Output	Disease label + localization + heatmap	Clinical decision support

7.1 Saved Artifacts

- **Classification Model:** cliniscan_best_cls_m3.pth — best ResNet50 checkpoint based on validation F1.
- **Detection Model:** yolo_m3/weights/best.pt — best YOLOv11s checkpoint based on mAP.
- **Class Names:** class_names.json — mapping of class IDs 0–13 to disease category names.
- **Dataset Config:** data_m3.yaml — YOLO dataset YAML with correct train/val paths and 14 class names.

8. CHALLENGES FACED

- **Class Imbalance:** Highly imbalanced class distributions (e.g., Aortic enlargement and Cardiomegaly dominate) remain a challenge. Label smoothing partially mitigates overconfidence on majority classes.
- **GPU Memory Constraints:** Reducing YOLO image size from 1024 to 640 allowed larger batch sizes (8→16) within Kaggle GPU limits, improving training stability.
- **Single-Label Approximation:** The classification model still uses a single primary label per image despite the dataset being multi-label. Full multi-label classification is targeted for Milestone 4.
- **Validation Leakage (M2):** Milestone 2 used the training set as both train and val for YOLO. This was corrected in Milestone 3 with a proper 80/20 split, enabling unbiased evaluation.
- **Grad-CAM Stability:** Hook-based Grad-CAM implementation required careful tensor detachment to avoid GPU memory leaks during batch visualization.

9. REFERENCES

- VinDr-CXR Dataset — <https://physionet.org/content/vindr-cxr/1.0.0/>
- Ultralytics YOLOv11 Documentation — <https://docs.ultralytics.com>
- PyTorch Documentation — <https://pytorch.org/docs>
- Alumentations Documentation — <https://alumentations.ai/docs>
- Selvaraju et al. (2017). Grad-CAM: Visual Explanations from Deep Networks via Gradient-based Localization. ICCV 2017.
- He et al. (2016). Deep Residual Learning for Image Recognition. CVPR 2016.